

Channel coding theorem

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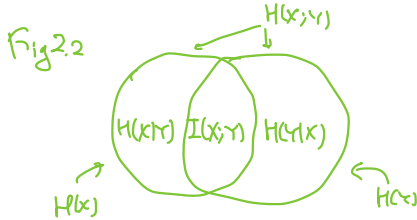
Claude Elwood Shannon
(April 30, 1916 – February 24, 2001)

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Def: A discrete channel is a system consisting of an input X , output Y , $p(y|x)$.

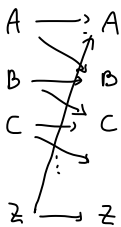
Information channel capacity: $C = \max_{p(x)} I(x; y)$



Thm 2.4.1.

$$I(X; Y) = H(X) - H(X|Y)$$

Ex) 2.1.3, Noisy Typewriter.



$$\begin{aligned} C &= \max [H(Y) - H(Y|X)] \\ &= \max H(Y) - 1 \\ &= \log_2 6 - 1 \\ &= \log_2 3 \end{aligned}$$

$$H(Y|X) = \log_2 2 = 1$$

Thm 2.6.4.

Thm 2.6.4. $H(X) \leq \log |X|$. equality holds iff X has a uniform disc. over X .

§ 1.5 definition for theorem.

Def. • A discrete channel: $(\mathcal{X}, p(y|x), \mathcal{Y})$

finite set $\rightarrow$ prob. mass fn.

• The n th extension of DMC: $\leftarrow$ discrete memoryless channel

$$(\mathcal{X}^n, p(y^n, x^n), \mathcal{Y}^n) \quad \text{where} \quad p(y_k | x^k, y^{k-1}) = p(y_k | x_k)$$

$\downarrow$
 $y^n = (y_1, y_2, \dots, y_n)$: vector notation.

• A channel without feedback if

$$p(x_k | x^{k-1}, y^{k-1}) = p(x_k | x^{k-1})$$

Def. An (M, n) code for $(\mathcal{X}, p(y|x), \mathcal{Y})$:

1. An index set: $\{1, 2, \dots, M\} =: \mathcal{I}^M$

codewords $\in$ codebook.

2. encoding fn: $x^n: \mathcal{I}^M \rightarrow \mathcal{X}^n$ $i \mapsto x^n(i)$

3. decoding fn.: $g: \mathcal{Y}^n \rightarrow \mathcal{I}^M$

Def. • Conditional prob. error. $\lambda_i = P(g(Y^n) \neq i | X^n = x^n(i))$

• The maximal prob. of error. $\lambda^{(n)} = \max_i \lambda_i$

• The average prob. of error. $p_e^{(n)} = \frac{1}{M} \sum_{i=1}^M \lambda_i$

- If the index W is chosen uniformly over $\mathcal{I}^M$,

$$p_e^{(n)} = P(W \neq g(Y^n))$$

Def. The rate R of (M, n) Code: $R = \frac{\log M}{n}$

• A rate R is achievable $\rightarrow 2^{nR} \in \mathbb{Z}$

if $\exists$ a seq. of $(2^{nR}, n)$ Code, s.t. $\lambda^{(n)} \xrightarrow{n \rightarrow \infty} 0$

• The capacity of a DMC is the supremum of all achievable rates.
(achievable capacity)

typical set

Def. The set $A_\varepsilon^{(n)}$ of jointly typical sequences $\{(x^n, y^n)\}$:

$$A_\varepsilon^{(n)} := \{(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n :$$

$$|-\frac{1}{n} \log p(x^n) - H(x)| < \varepsilon, |-\frac{1}{n} \log p(y^n) - H(y)| < \varepsilon, |-\frac{1}{n} \log p(x^n, y^n) - H(x, y)| < \varepsilon\}$$

Where

$$p(x^n, y^n) = \prod_{i=1}^n p(x_i, y_i)$$

$$\Downarrow$$

$$p(x_i) p(y_i)$$

Thm 7.6.1. (X^n, Y^n) : sequences drawn i.i.d. ad $p(x^n, y^n) = \prod_{i=1}^n p(x_i, y_i)$

1. $P((X^n, Y^n) \in A_\varepsilon^{(n)}) \rightarrow 1$ as $n \rightarrow \infty$

2. $|A_\varepsilon^{(n)}| \leq 2^{n(H(x, y) + \varepsilon)}$

3. If $(\tilde{x}^n, \tilde{y}^n) \sim p(x^n) p(y^n)$, then $P((\tilde{x}^n, \tilde{y}^n) \in A_\varepsilon^{(n)}) \leq 2^{-n(I(x, y) - 3\varepsilon)}$

For $n \gg 1$, $P((\tilde{x}^n, \tilde{y}^n) \in A_\varepsilon^{(n)}) \geq (1 - \varepsilon) 2^{-n(I(x, y) + 3\varepsilon)}$

Thm. (Channel Coding Theorem.) For a DMC,

" $R < C \Rightarrow \exists (2^{nR}, n)$ code with $d^{(n)} \rightarrow 0$ as $n \rightarrow \infty$ "

Conversely, if $\exists$ a seq. of $(2^{nR}, n)$ codes with $d^{(n)} \rightarrow 0$, $R \leq C$

pt. (Achievability) Fix $p(x)$. ^{distribution.} generate $(2^{nR}, n)$ code at random and

$$p(x^n) = \prod_{i=1}^n p(x_i)$$

Which gives a codebook:

$$C = \begin{bmatrix} x_1(1) & x_2(1) & \dots & x_n(1) \\ \vdots & \vdots & \ddots & \vdots \\ x_1(2^{nR}) & x_2(2^{nR}) & \dots & x_n(2^{nR}) \end{bmatrix} \rightarrow \text{the 1st merge goes to } n\text{-vector,}$$

Note that $P(C) = \prod_{\omega=1}^{2^{nR}} p(x^n(\omega)) = \prod_{\omega=1}^{2^{nR}} \prod_{i=1}^n p(x_i(\omega))$

seq. of events:

1. C is generated and $p(x)$.

2. C is revealed to encoder and decoder
 $\downarrow$
 know $p(y|x)$

3. W is chosen and a uniform dist: $P(W=\omega) = 2^{-nR}$, $\omega=1, 2, \dots, 2^{nR}$.

4. $X^n(\omega)$ is sent.

5. decoder receives a seq. Y^n and $P(y^n|x^n(\omega)) = \prod_{i=1}^n p(y_i|x_i(\omega))$.

6. The receiver decodes the index:

$$\hat{w} = g(\gamma^n) \text{ if } \begin{cases} (x^n(\hat{w}), \gamma^n) \text{ is jointly typical.} \\ \nexists w' (\neq \hat{w}) \text{ s.t. } (x^n(w'), \gamma^n) \in A_{\epsilon}^{(n)}. \end{cases}$$

error, when no such $\hat{w}$ exists or there is more than one such.

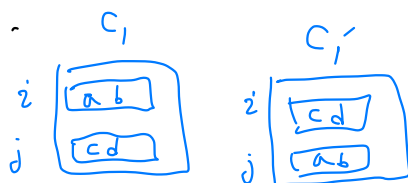
7. Decoding error if $\hat{w} \neq w$. $\mathcal{E} = \{\hat{w} \neq w\}$.

(Study of prob. of error)

Calculate the ave. prob. of error, averaged over all codes.

$$\begin{aligned} P(\mathcal{E}) &= \sum_C P(C) P_{\mathcal{E}}^{(n)}(C) & \underline{P_{\mathcal{E}}^{(n)} = P(w \neq g(\gamma^n))} \\ &= \sum_C P(C) \frac{1}{2^{nR}} \sum_{w=1}^{2^{nR}} \lambda_w(C) \\ &= \frac{1}{2^{nR}} \sum_{w=1}^{2^{nR}} \sum_C P(C) \lambda_w(C). \end{aligned}$$

By symmetry of code construction, $\sum_C P(C) \lambda_w(C)$ does not depend on w .



WOLG ass. $w=1$ was sent.

$$P(\mathcal{E}) = \frac{1}{2^{nR}} \sum_{w=1}^{2^{nR}} \sum_c P(c) \lambda_1(c).$$

$$= \sum_c P(c) \lambda_1(c) = P(\mathcal{E} | W=1).$$

Define $E_i = \{(X^n(i), Y^n) \in A_{\epsilon}^n\}$ for $i = \{1, 2, \dots, 2^{nR}\}$

$W=1$

- E_1^c $(X^n(1), Y^n)$ is not jointly typical.

An error occurs:

- $E_2 \cup \dots \cup E_{2^{nR}}$ Wrong codeword is jointly typical.

Y^n is the result of sending the first code word $X^n(1)$.

$$P(\mathcal{E} | W=1) = P(E_1^c \cup E_2 \cup \dots \cup E_{2^{nR}})$$

$$\leq P(E_1^c) + \sum_{i=2}^{2^{nR}} P(E_i)$$

$$P(A \cup B) \leq P(A) + P(B)$$

$$\leq \underbrace{\epsilon}_{(1)} + \underbrace{\sum_{i=2}^{2^{nR}} 2^{-n(I(X;Y) - 3\epsilon)}}_{(2)} \text{ by joint AEP.}$$

(3) $X^n(1)$ and $X^n(i)$: indep. for $i \neq 1$.

$$= \epsilon + (2^{nR} - 1) \cdot 2^{-n(I(X;Y) - 3\epsilon)}$$

$$\leq \epsilon + 2^{-n(I(X;Y) - 3\epsilon - R)} \leq 2\epsilon.$$

(if $R < I(X;Y) - 3\epsilon$ and $n \gg 1$.)

$\therefore$ For $R < I(X;Y)$, $\forall \epsilon, \exists n$ s.t. $P(\mathcal{E}) \leq 2\epsilon$.

(Finding a codebook we want)

1. Choose $p(x)$ that maximize $I(X;Y)$, and call it $p^*(x)$.

$$R < I(X;Y) \Rightarrow R < C = \max_{p(x)} I(X;Y)$$

2. $\exists C^*$ s.t. $P_e^{(n)}(C^*) \leq 2\varepsilon$. ↔ $P(\varepsilon) = \sum_c P(c) P_e^{(n)}$

3. Aim: $\lambda^{(n)}(C^*) \rightarrow 0$

$$2\varepsilon \geq \underbrace{\frac{1}{2^{nR}} \sum_{i=1}^{2^{nR}} \lambda_i(C^*)}_{\lambda^*} = \frac{1}{2^{nR}} \left(\sum_{\lambda_i \leq \lambda^*} \lambda_i + \sum_{\lambda_i > \lambda^*} \lambda_i \right)$$

↑ best half ↑ worst half.

For best half, $\lambda_i \leq 4\varepsilon$ for all i in best half.

If not $\lambda^* \geq 4\varepsilon$. • $\frac{1}{2^{nR}} \sum_{\lambda_i < \lambda^*} \lambda_i > \frac{1}{2} \cdot 4\varepsilon = 2\varepsilon$ *

Throw away the worst half of codeword in C^* .

And take 2^{nR-1} codewords (best half of C^*)

New Rate $\frac{1}{n} \cdot \log(2^{nR-1}) = R - \frac{1}{n}$

and $\lambda^{(n)} = \max \lambda_i \leq 4\varepsilon$

///

(Converse) "if $\exists$ a seq. of $(2^{nR}, n)$ codes with $d^{(n)} \rightarrow 0$, $R \leq C$ "

(special case)

Supp. $p_e^{(n)} = 0$, Ass: W : uniformly dist. over $\{1, 2, \dots, 2^{nR}\}$.

$$nR \stackrel{?}{=} H(W)$$

$$0 (\because p_e^{(n)} = 0; \hat{W} \stackrel{?}{=} g(Y^n))$$

$$\stackrel{\text{Fig 2.2}}{=} H(W|Y^n) + I(W; Y^n)$$

$$\leq I(X^n; Y^n)$$

: Data processing - $W \rightarrow X^n \rightarrow Y^n$.

$$\leq \sum_{i=1}^n I(X_i; Y_i)$$

1.9.2.

$$\leq nC$$

def. of C

$$\therefore R \leq C.$$

7.9.1 Lemma (Fano's inequality)

For a DMC, W : uniformly distributed. $H(X^n|Y^n) \leq 1 + p_e^{(n)} \cdot n \cdot R$

7.9.2. For DMC, $I(X^n; Y^n) \leq nC$.

$$\text{pf. } I(X^n; Y^n) = H(Y^n) - H(Y^n|X^n)$$

$$= H(Y^n) - \sum H(Y_i | Y_1, \dots, Y_{i-1}, X^n)$$

$$= H(Y^n) - \sum H(Y_i | X_i)$$

$\leftarrow$ DMC.

$$\leq \sum H(Y_i) - \sum H(Y_i | X_i)$$

$$= \sum I(X_i; Y_i) \leq nC.$$

* Converse of the coding theorem.

" if $\exists$ a seq. of $(2^{nR}, n)$ codes with $d^{(n)} \rightarrow 0$, $R \leq C$ "

pf. Since $p_e^{(n)} \leq \lambda^{(n)}$, $\lambda^{(n)} \rightarrow 0$ implies $p_e^{(n)} \rightarrow 0$.

Ass. w : uniform dist with $p_0^{(n)} = P(\hat{w} \neq w)$.

$$nR = H(w) = H(w|Y^n) + I(w; Y^n)$$

$$\leq H(w|Y^n) + I(X^n(w); Y^n)$$

$$w \rightarrow X^n \rightarrow Y^n$$

$$\leq 1 + p_n^{(e)} nR + I(X^n; Y^n)$$

$$\leq 1 + p_n^{(e)} nR + nC$$

Dividing by n , $R \leq \frac{1}{n} + p_n^{(e)} R + C$.

As $n \rightarrow \infty$, $R \leq C$.

///

Equity in the converse

We consider the special case $p_n^{(e)} = 0$: zero error code

$$nR \stackrel{?}{=} H(W)$$

$$\stackrel{?}{=} H(W|\hat{W}) + I(W;\hat{W})$$

(∵ $p_n^{(e)} = 0$)

Fig 2.2. X^n must be distinct.

$W \rightarrow \text{code}$

$$\stackrel{?}{\leq} I(X^n(W); Y^n)$$

data processing ineq. : $W \rightarrow X^n \rightarrow Y^n \rightarrow \hat{W}$.

$$= H(Y^n) - H(Y^n | X^n)$$

$$I(Y^n; X^n(W) | W) = 0 = I(X^n; Y^n | \hat{W})$$

$$= H(Y^n) - \sum_{i=1}^n H(Y_i | X_i) \quad \leftarrow \text{DMC}$$

$$\leq \sum_{i=1}^n H(Y_i) - \sum_{i=1}^n H(Y_i | X_i) \quad \leftarrow Y_i: \text{indep.}$$

$$= \sum_{i=1}^n I(X_i; Y_i) \quad \leftarrow \text{by def.}$$

$$\leq nC$$

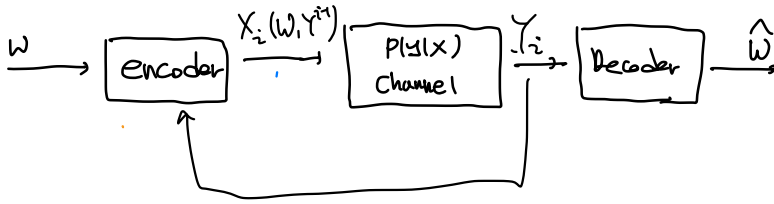
$\leftarrow \text{by def.}$

$$X_i \sim p^*(x_i)$$

$$\parallel$$

$$\arg \max I(X; Y)$$

§ 11.12 Feedback capacity



- $(2^{nR}, n)$ feedback code: a sequence of mappings $x_i(w, y^{i-1})$,
seq of decodg fns, $g: y^n \rightarrow \{1, 2, \dots, 2^{nR}\}$

The capacity with feedback, C_{FB} , of a DMC

: a supremum of all rates achievable by feedback codes.

$$1.12.1. \quad C_{FB} = C = \max_{p(x)} I(X; Y)$$

pt. A non feedback code is a special code of a feedback code

$$C_{FB} \geq C.$$

$$\text{Aim: } C_{FB} \leq C.$$

$$nR = H(W) = H(W|\hat{W}) + I(W; \hat{W})$$

$$(*) \quad \leq (1 + P_e^{(n)})nR + I(W; \hat{W})$$

$$\leq 1 + P_e^{(n)}nR + \underbrace{I(W; Y^n)}_{\text{yellow highlight}} \leftarrow W \rightarrow Y^n \rightarrow \hat{W}.$$

$$I(\omega; Y^n) = H(Y^n) - H(Y^n | \omega)$$

$$= H(Y^n) - \sum_{i=1}^n H(Y_i | Y_1, Y_2, \dots, Y_{i-1}, \omega)$$

$$\leq H(Y^n) - \sum_{i=1}^n H(Y_i | \underbrace{Y_1, Y_2, \dots, Y_{i-1}}_{X_i}, \omega, X_i) \quad \text{def } X_i = X_i(\omega, Y^{i-1})$$

$$= H(Y^n) - \sum_{i=1}^n H(Y_i | X_i)$$

DMC

$$\leq \sum_{i=1}^n H(Y_i) - \sum_{i=1}^n H(Y_i | X_i)$$

Chann. model + 2.6.1

$$= \sum_{i=1}^n I(X_i; Y_i)$$

def

$$H(X|Y) \leq H(X)$$

$$\leq nC.$$

def

$$\text{from (*)}, nR \leq P_e^{(n)} nR + 1 + nC.$$

$$\text{Dividing } n, \quad n \rightarrow \infty, \quad R \leq C.$$

$$C_{FB} \leq C.$$

//

§ 1.3: Source channel separation theorem.

Two stage method.

$$\left(\begin{array}{l} \text{ch T} \quad 5.4.2. \rightarrow R_s > H \\ \text{ch II} \quad 1.1.1' \rightarrow R_s \leq C. \end{array} \right) \rightarrow H(U) < C.$$

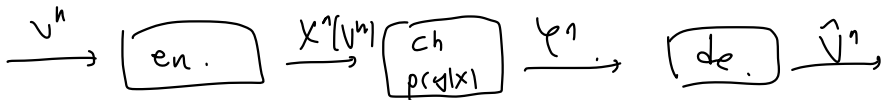
$$\begin{aligned} \text{§ 4.2} \quad \frac{H(x_1, \dots, x_n)}{n} &\leq L_n^x = \underbrace{\frac{n \log M}{n}}_{H(U)} \leq \frac{H(x_1, \dots, x_n)}{n} + \frac{1}{n} \\ &\quad \text{H(U)} \quad \text{H} \leq R \\ &\quad k = \log M \rightarrow M = 2^k, \quad k = n L_n^x, \quad \underline{H(U) \leq C}. \end{aligned}$$

Source-channel separation thm:

two-stage process is as good as any other method.

Source V. process produced by V satisfies AEP. (ex) i.i.d. ← stationary.

$$\text{Def. Prob. of error: } P(V \neq \hat{V}) = \sum_{y^n} \sum_{x^n} p(y^n) p(x^n | y^n) I(y^n \neq V^n)$$



Thm (Joint Source-Channel coding thm): DMC.

If $V_1, V_2, \dots, V_n$ is a finite alphabet stochastic process that satisfies the AEP and $\underline{H(U) < C}$,

$\exists$ source-channel code with $\underline{Pr(\hat{V} \neq V)} \rightarrow 0$

Conversely, for any stationary stochastic process, if $H(U) > C$, the probability of error is bounded away from zero.

pt: Achievability ($\rightarrow$) $\text{Aim: } R < C \Rightarrow p_e^{(n)} \rightarrow 0$
 $\uparrow H(U)$ replace.

AEP assumption on the process: $|A_\epsilon^{(n)}| < 2^{n(H(U) + \epsilon)}$ (thm 3.1.2 - 3)

We index all seq. in $A_\epsilon^{(n)}$ with $n(H(U) + \epsilon)$ bits
 $\downarrow$
 compress.

$$3.1.2-2: P(A_{\varepsilon}^{(n)}) > 1 - \varepsilon.$$

From ch coding thm, we can transmit the indices with the arbitrary small prob

$$\text{of error if } H(U) + \varepsilon = R < C$$

Can reconstruct V^n to agree with the transmitted sequence with high prob.

(why, input size $= n$ = output size)

Precisely,

$$\begin{aligned} P(V^n \neq \hat{V}^n) &\leq P(V^n \notin A_{\varepsilon}^{(n)}) + P(g(\gamma^n) \neq V^n | V^n \in A_{\varepsilon}^{(n)}) \\ &\leq \varepsilon + \varepsilon = 2\varepsilon \end{aligned}$$

$$\therefore H(U) < C.$$

Converse: " $P(V^n \neq \hat{V}^n) \rightarrow 0 \Rightarrow H(U) \leq C$ "

Given the following encoder and decoder

$$x^n(u^n) : U^n \rightarrow \mathcal{X}^n$$

$$g_n(\gamma^n) : \mathcal{Y}^n \rightarrow U^n$$

By Fano's ineq.

$$H(V^n | \hat{V}^n) \leq (1 + P_r(\hat{V}^n \neq V^n) \log |U^n|) = (1 + P_r(\hat{V}^n \neq V^n) n \log |U|)$$

||
|U|^n (only seq.)

$$H(U) \leq \frac{H(X_1, \dots, X_n)}{n} \quad \text{ch 4. } H(U) = \lim_{n \rightarrow \infty} \frac{H(U_1, \dots, U_n)}{n}, \text{ \#4.6 \# dec. seq.}$$

$$= \frac{H(V^n)}{n}$$

$$= \frac{1}{n} H(V^n | \hat{V}^n) + \frac{1}{n} I(V^n; \hat{V}^n)$$

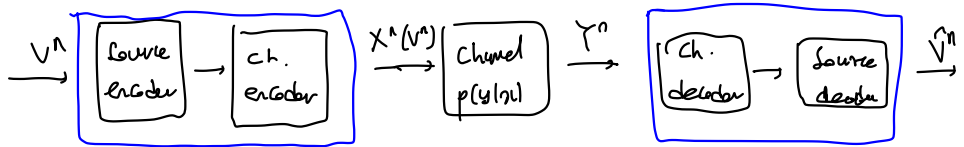
$$\leq \frac{1}{n} (1 + P(V^n \neq \hat{V}^n) n \log |U|) + \frac{1}{n} I(V^n; \hat{V}^n) \quad \text{Fano}$$

$$\leq \frac{1}{n} (1 + P(V^n \neq \hat{V}^n) n \log |U|) + \frac{1}{n} I(x^n; \gamma^n) \quad \text{Data processing}$$

$$\leq \frac{1}{n} (1 + P(V^n \neq \hat{V}^n) n \log |U|) + C \quad \text{Memoryless}$$

$$\text{As } n \rightarrow \infty, \therefore H(U) \leq C$$

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Summary, Data Compression thm: a consequence of the AEP.

Data transmission thm: a consequence of the joint AEP.